

Space-borne GPS remote sensing for atmospheric research

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ABSTRACT

The Global Positioning System (GPS) is based on a constellation of 24 transmitter satellites orbiting the earth at ~ 21,000 km altitude. The original goal of the GPS was to provide global and all-weather precision positioning and navigation for the military. Since this original concept was developed, several civilian applications have been conceived^{1,2} that are making use of these satellites.

GPS/MET³ is one such application. GPS/MET is sponsored by NSF, FAA, NOAA, and NASA. The goal of GPS/MET is to demonstrate the feasibility of recovering atmospheric temperature profiles from occulting radio signals from one of the 24 GPS transmitters.

On April 3, 1995, a small radio receiver was launched into a 750 km low-earth orbit and 70° inclination. As this receiver orbits, occultations occur when the radio link between any one of the 24 GPS transmitters and the low-earth orbiting (LEO) receiver progressively descends or ascends through the earth's atmosphere. With the current constellation of GPS transmitters, approximately 500 such occultations occur in each 24 hour-period per LEO receiver.

Several hundred occultations have been analyzed to date, where some type of confirmational data has been available (*i.e.*, radiosonde, satellite, numerical analysis gridded data). In this paper, we present a brief outline of the method followed by a few typical temperature soundings that have been obtained.

1. DISCUSSION OF TECHNIQUE

The GPS/MET experiment is based on the occultation method⁴ first developed at Stanford University and employed for studies of planetary atmospheres by the Jet Propulsion Laboratory (JPL).⁵⁻¹⁰ The technique is based upon the fact that, as a radio ray traverses nearly tangentially through the atmosphere, due to index of refraction, μ , changes, the ray is slowed down and bent. In particular, the bending is due to gradients of the index normal to the direction of propagation of the beam. With the assumption of spherical symmetry of the index distribution, the bending of the beam is directly related to the radial gradient of the index, and this gradient may therefore be determined from a measurement of the bending angle.

Figure 1 shows schematically a ray from a GPS transmitter to a LEO receiver, being bent through a bending angle, α , by the earth's atmosphere. In this figure, the radius of closest approach of the ray to the center of the earth is r , and a is the impact parameter, defined as the perpendicular distance from the center of the earth to the tangential extension of the received and/or transmitted rays. Note that, under the assumption of spherical symmetry, these impact parameters are equal.

From the measured signal at the LEO, the frequency of the received ray may be determined. Due to the motions of the transmitter and receiver, this frequency undergoes a Doppler shift. This shift will be different than the Doppler shift which would result if the ray travelled a straight path from transmitter to receiver, due to the different angle between the transmitted ray and the transmitter velocity vector, and the received ray and the receiver velocity vector, resulting from the bending. The Doppler shift, therefore, is a measure of the bending angle, α .

Once the bending angle has been determined, the index of refraction profile, μ , may be retrieved from the Abel transform, with the assumption of spherical symmetry. Thus,

$$\mu(r_m) = \exp \left[\frac{1}{\pi} \int_{a=a_m}^{\infty} \ln \left\{ \frac{a}{a_m} + \left[\left(\frac{a}{a_m} \right)^2 - 1 \right]^{\frac{1}{2}} \frac{d\alpha}{da} da \right\} \right] \quad (1)$$

Here r_m is the radius of closest approach of the m^{th} ray, a_m is the corresponding impact parameter, and α is the measured bending angle. By solving Eq. (1) for rays which penetrate successively lower and lower into the atmosphere (i.e., successively smaller and smaller values of r_m), the index of refraction profile, $\mu(r_m)$, may be determined. If we define the refractivity, N , as

$$N = (\mu - 1) \times 10^6 \quad (2)$$

then the refractivity is related to the atmospheric density, ρ , in kg/m^3 by the expression

$$N = 77.60 \rho R + 3.730 \times 10^5 \left(\frac{e}{T^2} \right) \quad (3)$$

where R is the gas constant for dry air, T is the temperature in degrees Kelvin, and e is the vapor pressure in millibars. In regions where water vapor can be ignored (i.e., usually above 5–8 km), the last term on the right-hand side of Eq. (3) may be neglected and the atmospheric density profile may be determined from the measured profile of N . The pressure profile, p , as a function of altitude, z , may then be determined from the hydrostatic equation:

$$dp = -\rho g dz \quad (4)$$

where g is the gravitational acceleration, and z is height above the surface. By starting at a high enough level, such that the initial pressure is known to within fractions of a millibar, Eq. (4) may be integrated level by level to obtain $p(z)$, the pressure as a function of height. Finally, from the equation of state

$$p = \rho RT, \quad (5)$$

the temperature profile may be determined from the previously determined profiles of p and ρ .

It should be noted that, when water vapor becomes significant, such that the second term in Eq. (3) is no longer negligible, the dry air assumption will lead to erroneously low temperatures.

2. SOME RESULTS

Several hundred soundings have been analyzed to date. Figures 2–5 present 4 typical temperature retrievals. They are quite representative of the entire collection.

Figure 2 represents a retrieval from northern Minnesota with 2 radiosonde comparison soundings (i.e., RS), two CLASS (cross-chain Loran Sounding System) systems (CL), and one National Meteorological Center (NMC) gridded analysis sounding. The excellent agreement among all these sets of data is obvious. Note that for this case, the GPS data terminated at slightly below 5 km. In most instances, reliable results have not been obtained to the surface due to several factors. Some of these factors may be eliminated, which will enable a certain percentage of the soundings to extend to, or near to, the surface. However, due to such propagation effects as multi-path, ground reflection, focusing and defocusing, etc., many soundings will still not be retrievable to the surface. Of note in this and succeeding soundings is the accurate recovery of the tropopause. One of the most noteworthy features of the GPS/MET technique is the accuracy with which the tropopause height and structure are reproduced.

Figure 3 is a retrieval from the coastal region of northeastern Siberia. Once again there is general overall agreement among all the data sets. The effects of tropospheric water vapor on the "dry air" retrievals may be clearly seen below about 5 km. The effect of water vapor can readily be shown to cause a "dry air" retrieval to produce a temperature which is too cold. Clearly the GPS retrieval is producing values which are 10–15°C colder than the other sets of data near the surface. This difference between the GPS/MET "dry air" temperature and the actual temperature may be used to retrieve the water vapor. If the temperature is

known to within about $\pm 2^{\circ}\text{C}$, the relative humidity may be retrieved by this method to within about $\pm 10\%$.

Figure 4 is another retrieval from eastern Asia. For this case, a tremendous amount of comparison data was available. The GPS/MET retrieval seems to represent a reasonable average of all these data.

Finally, Figure 5 is a sounding from the southern Australia coast. The radiosonde, NMC, and GPS/MET data are all in excellent agreement. Note that at about 3.5 km the GPS/MET sounding clearly goes bad. These erroneous values are probably due to multi-path effects in this case. The bad values are easily identifiable, as they are in most cases, and have no effect on the retrievals above the level where they go bad.

From the early results, certain things have become quite apparent. Firstly, GPS/MET provides accurate temperature retrievals from about 5 km to at least 25–30 km. The accuracies above these levels are not known due to the lack of confirmational data. Some early comparisons with HALOE (Halogen Occultation Experiment) retrievals have indicated agreements to within about $\pm 2^{\circ}\text{C}$ above about 30 km between the two sets of data. Above about 40 km, the GPS/MET data become somewhat noisy. We are currently working on methods to smooth this data and extend the height of useful retrievals to perhaps 50 km. Also, techniques to extend the useful data to lower altitudes are currently being investigated. Some of these problems involve the software package in the receiver and are presently being addressed. Other issues involving multi-path effects, ground reflection, separation of temperature and water vapor, etc., are topics for future investigation.

3. REFERENCES

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THE OCCULTATION MEASUREMENT

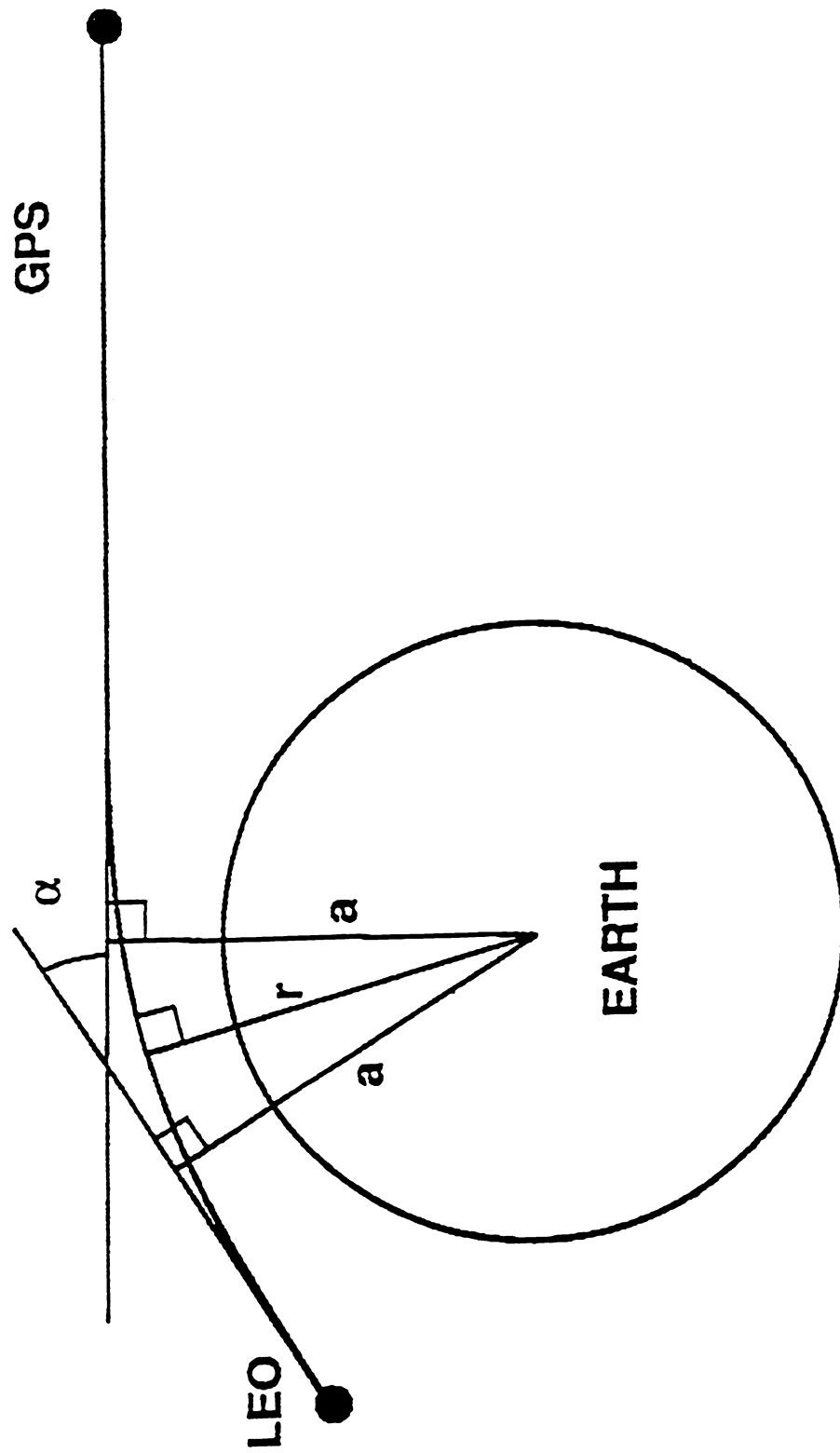


Figure 1

Day 124 Occ #079

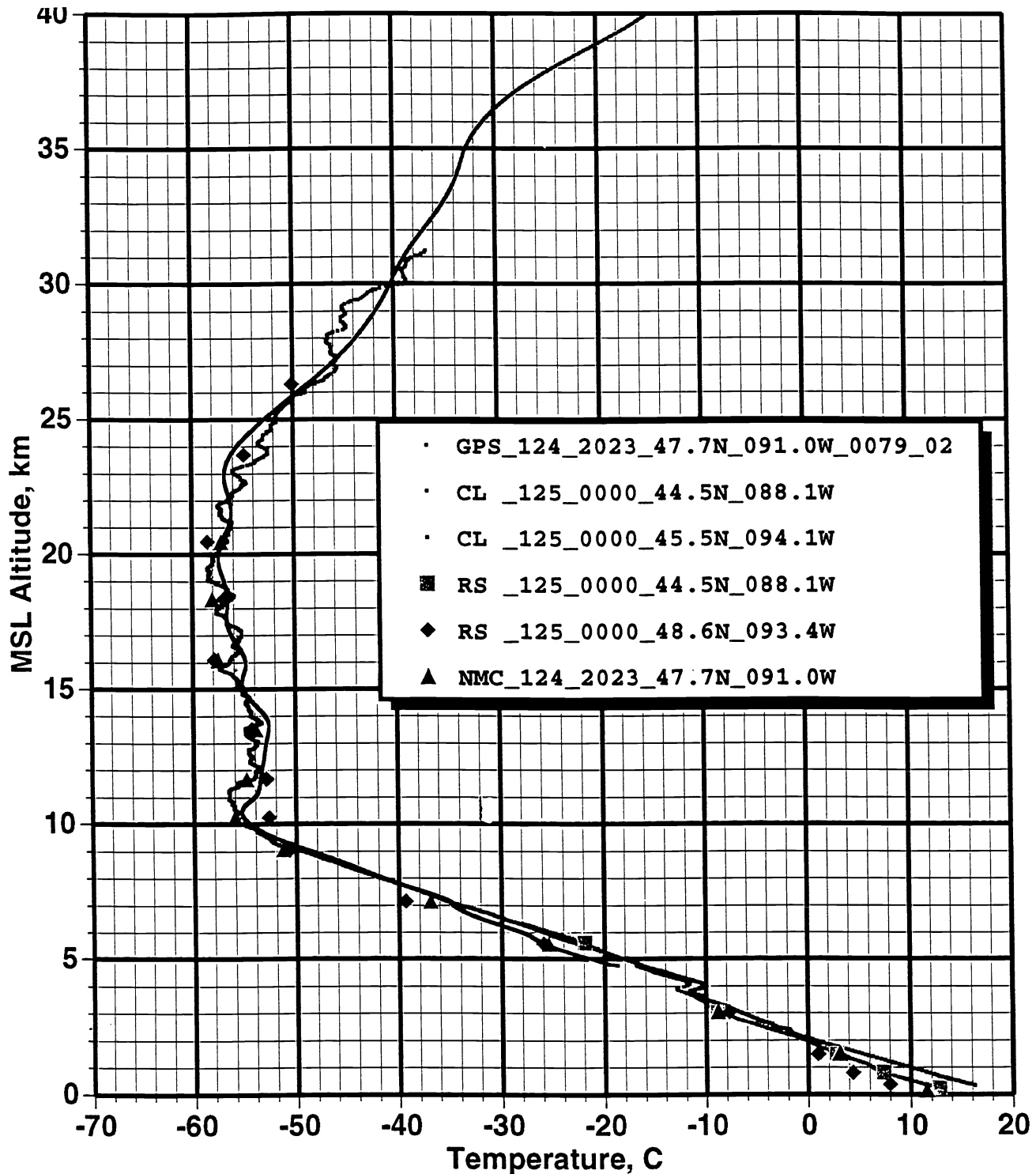


Figure 2

Day 125 Occ #028

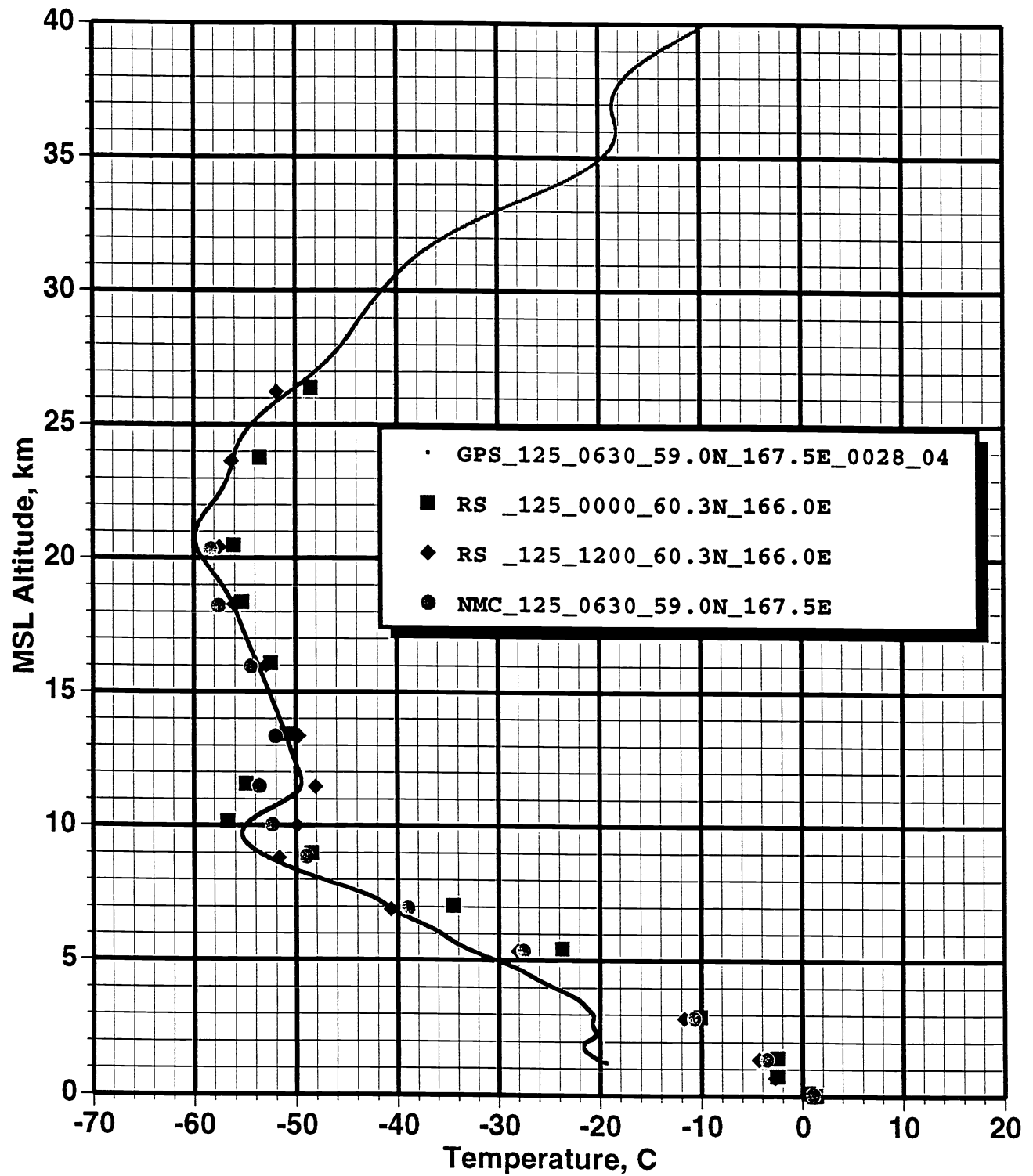


Figure 3

Day 125 Occ #020

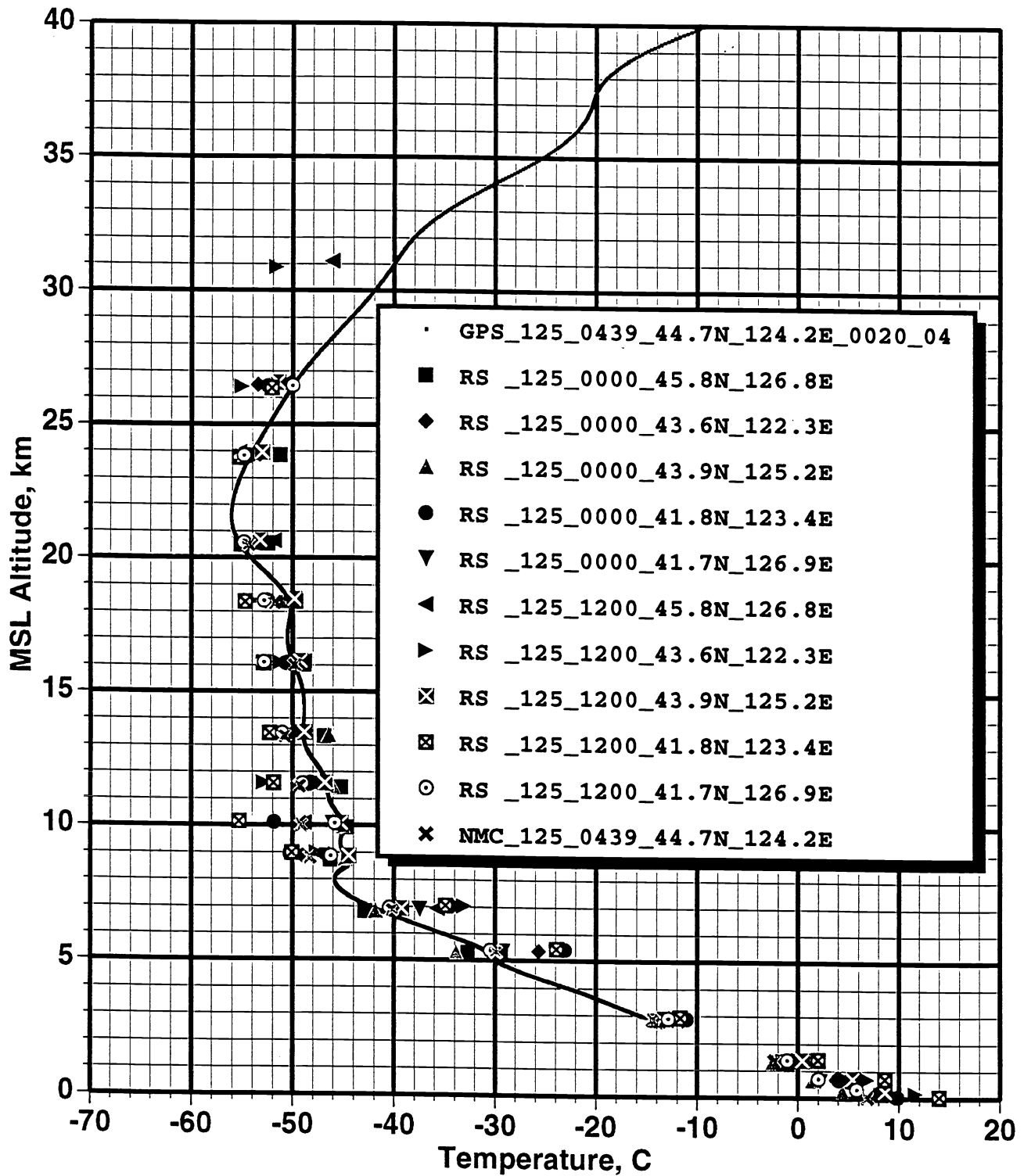


Figure 4

Day 124 Occ #057

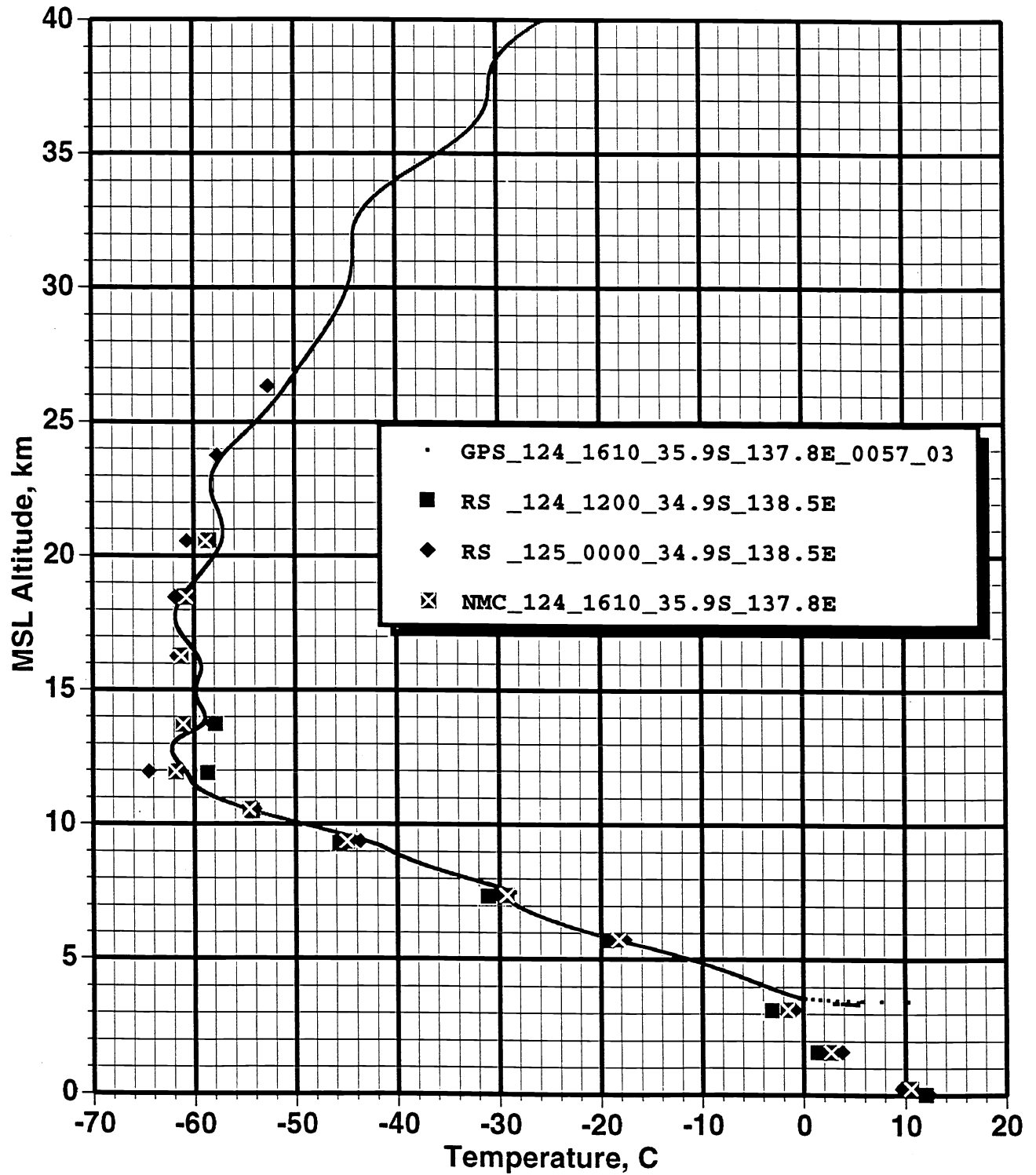


Figure 5